

Exalples

Wei December 26, 2025

Proposition 0.0.1. *Suppose that $\lambda \in X$, $\alpha_i \in \Phi_0^+$, $1 \leq i \leq k$, and $\beta, \gamma \in \Phi_1^+$. Let $w = s_{\alpha_k} s_{\alpha_{k-1}} \cdots s_{\alpha_1} \in \mathcal{W}$.*

- (1) *Suppose that $\langle \lambda, \alpha_1^\vee \rangle < 0$. Then $(P_\lambda : M_{s_{\alpha_1} \lambda}) > 0$.*
- (2) *Suppose that $\langle s_{\alpha_{i-1}} \cdots s_{\alpha_1} \lambda, \alpha_i^\vee \rangle < 0$ for all $i \in \{1, 2, \dots, k\}$. Then $(P_\lambda : M_{w\lambda}) > 0$.*
- (3) *Suppose that $(\lambda, \beta) = 0$. Then $(P_\lambda : M_{\lambda+\beta}) > 0$.*
- (4) *Suppose that $(\lambda, \beta) = 0$ and $\langle s_{\alpha_{i-1}} \cdots s_{\alpha_1}(\lambda + \beta), \alpha_i^\vee \rangle < 0$ for all $i \in \{1, 2, \dots, k\}$. Then $(P_\lambda : M_{w(\lambda+\beta)}) > 0$.*
- (5) *Suppose that $(\lambda, \beta) = (\lambda + \beta, \gamma) = 0$ and $\text{ht}(\beta) < \text{ht}(\gamma)$. Then $(P_\lambda : M_{\lambda+\beta+\gamma}) > 0$.*
- (6) *Suppose that $(\lambda, \beta) = (\lambda + \beta, \gamma) = 0$, $\text{ht}(\beta) < \text{ht}(\gamma)$, and $\langle s_{\alpha_{i-1}} \cdots s_{\alpha_1}(\lambda + \beta + \gamma), \alpha_i^\vee \rangle < 0$ for all $i \in \{1, 2, \dots, k\}$. Then $(P_\lambda : M_{w(\lambda+\beta+\gamma)}) > 0$.*

Lemma 0.0.1. *If $\lambda - \rho$ is the lowest weight in a standard filtration of a projective object P , then P_λ is a direct summand of P .*

1 $\mathfrak{gl}(2|2)$

1.1 degree of atypicality = 1

Theorem 1.1. *Let $a, b, c \in \mathbb{Z}$ and $\lambda = (a, b|c, b)$. The projective objects P_λ in $\mathfrak{B}_{a|c}$ have the following Verma flag formulae.*

Proof. The Cases are as follows:

- (1) $a > b > c$

$$(1.1) \quad a > b + 1$$

$$\text{Let } \mu := \lambda + (0, 1, 0, 0).$$

$$P_\mu = M_{a, b+1|c, b}$$

$$Pr_\lambda(P_\mu \otimes V_2) = M_{a, b|c, b} + M_{a, b+1|c, b+1}$$

All items belong to P_λ by [Proposition 0.0.1](#).

$$(1.2) \quad a = b + 1$$

$$\text{Let } \mu := \lambda + (1, 1, 0, 0).$$

$$P_\mu = M_{b+2, b+1|c, b}$$

$$Pr_\lambda(P_\mu \otimes V_6) = M_{b+1, b|c, b} + M_{b+1, b+1|c, b+1} + M_{b+2, b+1|c, b+2}$$

All items belong to P_λ by [Proposition 0.0.1](#).

- (2) $a = b > c$ Let $\mu := \lambda + (0, 0, 0, 1)$.

$$P_\mu = M_{a, a|c, a+1}$$

$$Pr_\lambda(P_\mu \otimes V_2) = M_{a+1, a|c, a+1} + M_{a, a+1|c, a+1} + M_{a, a|c, a} = M_{a, a|c, a} + \sum M_{a, a+1|c, a+1}$$

All items belong to P_λ by [Proposition 0.0.1](#).

- (3) $a > b = c$

(3.1) $a > b + 1$ Let $\mu := \lambda + (0, 1, 0, 0)$.

$$P_\mu = M_{a,b+1|b,b}$$

$$Pr_\lambda(P_\mu \otimes V_2) = +M_{a,b|b,b} + M_{a,b+1|b+1,b} + M_{a,b+1|b,b+1} = M_{a,b|b,b} + \sum M_{a,b+1|b+1,b}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(3.2) $a = b + 1$ Let $\mu := \lambda + (1, 1, 0, 0)$.

$$P_\mu = M_{b+2,b+1|b,b}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= M_{b+1,b|b,b} + M_{b+1,b+1|b+1,b} + M_{b+1,b+1|b,b+1} + M_{b+2,b+1|b+2,b} + M_{b+2,b+1|b,b+2} \\ &= M_{b+1,b|b,b} + \sum M_{b+1,b+1|b+1,b} + \sum M_{b+2,b+1|b+2,b} \end{aligned}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(4) $b > a > c$ Let $\mu := \lambda + (0, 1, 0, 0)$.

$$P_\mu = M_{a,b+1|c,b} + M_{b+1,a|c,b}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_2) &= M_{a,b|c,b} + M_{a,b+1|c,b+1} + M_{b,a|c,b} + M_{b+1,a|c,b+1} \\ &= \sum M_{a,b|c,b} + \sum M_{a,b+1|c,b+1} \end{aligned}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(5) $a > c > b$

(5.1) $c > b + 1$ Let $\mu := \lambda + (0, 1, 0, 0)$.

$$P_\mu = M_{a,b+1|c,b} + M_{a,b+1|b,c}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_2) &= M_{a,b|c,b} + M_{a,b+1|c,b+1} + M_{a,b|b,c} + M_{a,b+1|b+1,c} \\ &= \sum M_{a,b|c,b} + \sum M_{a,b+1|c,b+1} \end{aligned}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(5.2) $c = b + 1$ Let $\mu := \lambda + (0, 0, 0, 1)$.

$$P_\mu = M_{a,b|b+1,b+1}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_1) &= +M_{a,b+1|b+1,b+1} + M_{a,b|b,b+1} + M_{a,b|b+1,b} \\ &= \sum M_{a,b|b+1,b} \sum M_{a,b+1|b+1,b+1} \end{aligned}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(6) $a > c = b$ Already calculated!

(7) $c > a > b$

(7.1) $a > b + 1$ Let $\mu := \lambda + (0, 1, 0, 0)$.

$$P_\mu = M_{a,b+1|c,b} + M_{a,b+1|b,c}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_2) &= +M_{a,b|c,b} + M_{a,b+1|c,b+1} + M_{a,b|b,c} + M_{a,b+1|b+1,c} \\ &= \sum M_{a,b|c,b} + \sum M_{a,b+1|c,b+1} \end{aligned}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(7.2) $a = b + 1, c > a + 1$ Let $\mu := \lambda + (1, 1, 0, 0)$.

$$P_\mu = M_{b+2,b+1|c,b} + M_{b+2,b+1|b,c}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= M_{b+1,b|c,b} + M_{b+1,b+1|c,b+1} + M_{b+2,b+1|c,b+2} \\ &\quad + M_{b+1,b|b,c} + M_{b+1,b+1|b+1,c} + M_{b+2,b+1|b+2,c} \\ &= \sum M_{b+1,b|c,b} + \sum M_{b+1,b+1|c,b+1} + \sum M_{b+2,b+1|c,b+2} \end{aligned}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(7.3) $a = b + 1, c = a + 1$ Let $\mu := \lambda + (0, 0, 0, 2)$.

$$P_\mu = M_{b+1,b|b+2,b+2}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_5) &= M_{b+2,b+1|b+2,b+2} + M_{b+1,b+1|b+1,b+2} + M_{b+1,b+1|b+2,b+1} \\ &\quad + M_{b+1,b|b,b+2} + M_{b+1,b|b+2,b} \\ &= \sum M_{b+1,b|b+2,b} + \sum M_{b+1,b+1|b+2,b+1} + M_{b+2,b+1|b+2,b+2} \end{aligned}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(8) $c > a = b$

(8.1) $c > a + 1$ Let $\mu := \lambda + (0, 0, 0, 1)$.

$$P_\mu = M_{a,a|c,a+1} + M_{a,a|a+1,c}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_1) &= M_{a+1,a|c,a+1} + M_{a,a+1|c,a+1} + M_{a,a|c,a} \\ &\quad + M_{a+1,a|a+1,c} + M_{a,a+1|a+1,c} + M_{a,a|a,c} \\ &= \sum M_{a,a|c,a} + \sum M_{a,a+1|c,a+1} \end{aligned}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(8.2) $c = a + 1$ Let $\mu := \lambda + (0, 0, 0, 1)$.

$$P_\mu = M_{a,a|a+1,a+1}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_1) &= M_{a+1,a|a+1,a+1} + M_{a,a+1|a+1,a+1} + M_{a,a|a,a+1} + M_{a,a|a+1,a} \\ &= \sum M_{a,a|a+1,a} + \sum M_{a,a+1|a+1,a+1} \sum M_{a,a|c,a} + \sum M_{a,a+1|c,a+1} \end{aligned}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(9) $b > c > a$ Let $\mu := \lambda + (0, 1, 0, 0)$.

$$P_\mu = M_{a,b+1|c,b} + M_{b+1,a|c,b}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_2) &= M_{a,b|c,b} + M_{a,b+1|c,b+1} + M_{b,a|c,b} + M_{b+1,a|c,b+1} \\ &= \sum M_{a,b|c,b} + \sum M_{a,b+1|c,b+1} \end{aligned}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(10) $b = c > a$ Already calculated!

(11) $c > b > a$

(11.1) $c > b + 1$ Let $\mu := \lambda + (0, 1, 0, 0)$.

$$P_\mu = M_{a,b+1|c,b} + M_{a,b+1|b,c} + M_{b+1,a|c,b} + M_{b+1,a|b,c}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_2) &= M_{a,b|c,b} + M_{a,b+1|c,b+1} + M_{a,b|b,c} + M_{a,b+1|b+1,c} \\ &\quad + M_{b,a|c,b} + M_{b+1,a|c,b+1} + M_{b,a|b,c} + M_{b+1,a|b+1,c} \\ &= \sum M_{a,b|c,b} + \sum M_{a,b+1|c,b+1} \end{aligned}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(11.2) $c = b + 1$ Let $\mu := \lambda + (0, 0, 0, 1)$.

$$P_\mu = M_{a,b|b+1,b+1} + M_{b,a|b+1,b+1}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_1) &= M_{a,b+1|b+1,b+1} + M_{a,b|b,b+1} + M_{a,b|b+1,b} + M_{b+1,a|b+1,b+1} \\ &\quad + M_{b,a|b,b+1} + M_{b,a|b+1,b} = \sum M_{a,b|b+1,b} + \sum M_{a,b+1|b+1,b+1} \end{aligned}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(12) $c > b = a$ Already calculated!

□

Theorem 1.2. Let $a, b, c \in \mathbb{Z}$ and $\lambda = (a, b|b, c)$. The projective objects P_λ in $\mathfrak{B}_{a|c}$ have the following Verma flag formulae.

Proof. The Cases are as follows:

(1) $a > b > c$

(1.1) $a > b + 1$ Let $\mu := \lambda + (0, 1, 0, 0)$.

$$P_\mu = M_{a,b+1|b,c} + M_{a,b+1|c,b}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_2) &= M_{a,b|b,c} + M_{a,b+1|b+1,c} + M_{a,b|c,b} + M_{a,b+1|c,b+1} \\ &= \sum M_{a,b|b,c} + \sum M_{a,b+1|b+1,c} \end{aligned}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(1.2) $a = b > c$ Let $\mu := \lambda + (1, 1, 0, 0)$.

$$P_\mu = M_{b+2,b+1|b,c} + M_{b+2,b+1|c,b}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_6) &= M_{b+1,b|b,c} + M_{b+1,b+1|b+1,c} + M_{b+2,b+1|b+2,c} \\ &\quad + M_{b+1,b|c,b} + M_{b+1,b+1|c,b+1} + M_{b+2,b+1|c,b+2} \\ &= \sum M_{b+1,b|b,c} + \sum M_{b+1,b+1|b+1,c} + \sum M_{b+2,b+1|b+2,c} \end{aligned}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(2) $a = b > c$ Let $\mu := \lambda + (0, 0, 1, 0)$.

$$P_\mu = M_{a,a|a+1,c} + M_{a,a|c,a+1}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_1) &= +M_{a+1,a|a+1,c} + M_{a,a+1|a+1,c} + M_{a,a|a,c} + M_{a+1,a|c,a+1} + M_{a,a+1|c,a+1} + M_{a,a|c,a} \\ &= \sum M_{a,a|a,c} + \sum M_{a,a+1|a+1,c} \end{aligned}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(3) $a > b = c$ Already calculated!

(4) $b > a > c$ Let $\mu := \lambda + (0, 1, 0, 0)$.

$$P_\mu = M_{a,b+1|b,c} + M_{a,b+1|c,b} + M_{b+1,a|b,c} + M_{b+1,a|c,b}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_2) &= M_{a,b|b,c} + M_{a,b+1|b+1,c} + M_{a,b|c,b} + M_{a,b+1|c,b+1} \\ &\quad + M_{b,a|b,c} + M_{b+1,a|b+1,c} + M_{b,a|c,b} + M_{b+1,a|c,b+1} \\ &= \sum M_{a,b|b,c} + \sum M_{a,b+1|b+1,c} \end{aligned}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(5) $a > c > b$

(5.1) $c > b + 1$ Let $\mu := \lambda + (0, 1, 0, 0)$.

$$P_\mu = M_{a,b+1|b,c}$$

$$Pr_\lambda(P_\mu \otimes V_2) = M_{a,b|b,c} + M_{a,b+1|b+1,c}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(5.2) $c = b + 1, a > c + 1$ Let $\mu := \lambda + (0, 0, 1, 1)$.

$$P_\mu = M_{a,b|b+1,b+2} M_{a,b+1|b,c}$$

$$Pr_\lambda(P_\mu \otimes V_3) = M_{a,b|b,b+1} + M_{a,b+1|b+1,b+1} + M_{a,b+2|b+1,b+2}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(5.3) $c = b + 1, a = c + 1$ Let $\mu := \lambda + (1, 0, 0, 0)$.

$$P_\mu = M_{b+3,b|b,b+1} + M_{b+3,b+1|b+1,b+1} + M_{b+3,b+2|b+1,b+2} M_{a,b+1|b,c}$$

$$Pr_\lambda(P_\mu \otimes V_2) = M_{b+2,b|b,b+1} + M_{b+2,b+1|b+1,b+1} + M_{b+2,b+2|b+1,b+2} + M_{b+3,b+2|b+1,b+3}$$

By [Proposition 0.0.1](#) and [Lemma 0.0.1](#), all terms except $M_{b+3,b+2|b+1,b+3}$ must belong to P_λ . Since $M_{b+4,b+2|b+1,b+4}$ must appear in $P_{b+3,b+2|b+1,b+3}$ by [Proposition 0.0.1](#), the one remaining term cannot form a projective, it must also belong to P_λ .

(6) $a > c = b$ Already calculated!

(7) $c > a > b$

(7.1) $a > b + 1$ Let $\mu := \lambda + (0, 1, 0, 0)$.

$$P_\mu = M_{a,b+1|b,c}$$

$$Pr_\lambda(P_\mu \otimes V_2) = M_{a,b|b,c} + M_{a,b+1|b+1,c}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(7.2) $a = b + 1$ Let $\mu := \lambda + (1, 1, 0, 0)$.

$$P_\mu = M_{b+2,b+1|b,c}$$

$$Pr_\lambda(P_\mu \otimes V_6) = M_{b+1,b|b,c} + M_{b+1,b+1|b+1,c} + M_{b+2,b+1|b+2,c}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(8) $c > a = b$

(8.1) $c > b + 1$ Let $\mu := \lambda + (0, 0, 1, 0)$.

$$P_\mu = M_{a,a|a+1,c}$$

$$Pr_\lambda(P_\mu \otimes V_1) = M_{a+1,a|a+1,c} + M_{a,a+1|a+1,c} + M_{a,a|a,c} = M_{a,a|a,c} + \sum M_{a,a+1|a+1,c}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(8.2) $c = b + 1$ Let $\mu := \lambda + (0, 0, 1, 1)$.

$$P_\mu = M_{a,a|a+1,a+2}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_3) &= +M_{a,a|a,a+1} + M_{a+1,a|a+1,a+1} + M_{a,a+1|a+1,a+1} \\ &\quad + M_{a+2,a|a+1,a+2} + M_{a,a+2|a+1,a+2} \\ &= M_{a,a|a,a+1} + \sum M_{a,a+1|a+1,a+1} + \sum M_{a,a+2|a+1,a+2} \end{aligned}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(9) $b > c > a$ Let $\mu := \lambda + (0, 0, 1, 0)$.

$$P_\mu = M_{a,b|b+1,c} + M_{a,b|c,b+1} + M_{b,a|b+1,c} + M_{b,a|c,b+1}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_1) &= M_{a,b+1|b+1,c} + M_{a,b|b,c} + M_{a,b+1|c,b+1} + M_{a,b|c,b} \\ &\quad + M_{b+1,a|b+1,c} + M_{b,a|b,c} + M_{b+1,a|c,b+1} + M_{b,a|c,b} \\ &= \sum M_{a,b|b,c} + \sum M_{a,b+1|b+1,c} \end{aligned}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(10) $b = c > a$ Already calculated!

(11) $c > b > a$

(11.1) $c > b + 1$ Let $\mu := \lambda + (0, 0, 1, 0)$.

$$P_\mu = M_{a,b|b+1,c} + M_{b,a|b+1,c}$$

$$Pr_\lambda(P_\mu \otimes V_1) = M_{a,b+1|b+1,c} + M_{a,b|b,c} + M_{b+1,a|b+1,c} + M_{b,a|b,c} = \sum M_{a,b|b,c} + \sum M_{a,b+1|b+1,c}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(11.2) $c = b + 1$ Let $\mu := \lambda + (0, 0, 1, 1)$.

$$P_\mu = M_{a,b|b+1,b+2} + M_{b,a|b+1,b+2}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_3) &= +M_{a,b|b,b+1} + M_{a,b+1|b+1,b+1} + M_{a,b+2|b+1,b+2} + M_{b,a|b,b+1} \\ &\quad + M_{b+1,a|b+1,b+1} + M_{b+2,a|b+1,b+2} \\ &= \sum M_{a,b|b,b+1} + \sum M_{a,b+1|b+1,b+1} + \sum M_{a,b+2|b+1,b+2} \end{aligned}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(12) $a > b = a$ Already calculated!

□

Theorem 1.3. Let $a, b, c \in \mathbb{Z}$ and $\lambda = (a, b|c, a)$ with $a \neq b$. The projective objects P_λ in $\mathfrak{B}_{b|c}$ have the following Verma flag formulae.

Proof. The Cases are as follows:

(1) $a > b > c$ Let $\mu := \lambda + (0, 0, 0, 1)$.

$$P_\mu = M_{a,b|c,a+1}$$

$$Pr_\lambda(P_\mu \otimes V_1) = M_{a+1,b|c,a+1} + M_{a,b|c,a}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(2) $b > a > c$

(2.1) $b > a + 1$ Let $\mu := \lambda + (0, 0, 0, 1)$.

$$P_\mu = M_{a,b|c,a+1} + M_{b,a|c,a+1}$$

$$Pr_\lambda(P_\mu \otimes V_1) = M_{a+1,b|c,a+1} + M_{a,b|c,a} + M_{b,a+1|c,a+1} + M_{b,a|c,a} = \sum M_{a,b|c,a} + \sum M_{a+1,b|c,a+1}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(2.2) $b = a + 1$ Let $\mu := \lambda + (1, 0, 0, 0)$.

$$P_\mu = M_{a+1,a+1|c,a}$$

$$Pr_\lambda(P_\mu \otimes V_2) = M_{a,a+1|c,a} + M_{a+1,a|c,a} + M_{a+1,a+1|c,a+1} = \sum M_{a,a+1|c,a} + M_{a+1,a+1|c,a+1}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(3) $b > a = c$

(3.1) $b > a + 1$ Let $\mu := \lambda + (1, 0, 0, 0)$.

$$P_\mu = M_{a+1,b|a,a} + M_{b,a+1|a,a}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_2) &= M_{a,b|a,a} + M_{a+1,b|a+1,a} + M_{a+1,b|a,a+1} + M_{b,a|a,a} + M_{b,a+1|a+1,a} + M_{b,a+1|a,a+1} \\ &= \sum M_{a,b|a,a} + \sum M_{a+1,b|a+1,a} \end{aligned}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(3.2) $b = a + 1$ Let $\mu := \lambda + (1, 0, 0, 0)$.

$$P_\mu = M_{a+1,a+1|a,a}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_2) &= M_{a,a+1|a,a} + M_{a+1,a|a,a} + M_{a+1,a+1|a+1,a} + M_{a+1,a+1|a,a+1} \\ &= \sum M_{a,a+1|a,a} + \sum M_{a+1,a+1|a+1,a} \end{aligned}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(4) $a > c > b$ Let $\mu := \lambda + (1, 0, 0, 0)$.

$$P_\mu = M_{a+1,b|c,a}$$

$$Pr_\lambda(P_\mu \otimes V_2) = M_{a,b|c,a} + M_{a+1,b|c,a+1}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(5) $a = c > b$ Let $\mu := \lambda + (1, 0, 0, 0)$.

$$P_\mu = M_{a+1,b|a,a}$$

$$Pr_\lambda(P_\mu \otimes V_2) = +M_{a,b|a,a} + M_{a+1,b|a+1,a} + M_{a+1,b|a,a+1} = M_{a,b|a,a} + \sum M_{a+1,b|a+1,a}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(6) $c > a > b$

(6.1) $c > a + 1$ Let $\mu := \lambda + (1, 0, 0, 0)$.

$$P_\mu = M_{a+1,b|c,a} + M_{a+1,b|a,c}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_2) &= M_{a,b|c,a} + M_{a+1,b|c,a+1} + M_{a,b|a,c} + M_{a+1,b|a+1,c} \\ &= \sum M_{a,b|c,a} + \sum M_{a+1,b|c,a+1} \end{aligned}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(6.2) $c = a + 1$ Let $\mu := \lambda + (0, -1, 0, 1)$.

$$P_\mu = M_{a,b-1|a+1,a+1}$$

$$Pr_\lambda(P_\mu \otimes V_5) = M_{a+1,b|a+1,a+1} + M_{a,b|a,a+1} + M_{a,b|a+1,a} = \sum M_{a,b|a+1,a} + M_{a+1,b|a+1,a+1}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(7) $c = a > b$ Let $\mu := \lambda + (1, 0, 0, 0)$.

$$P_\mu = M_{a+1,b|a,a}$$

$$Pr_\lambda(P_\mu \otimes V_2) = M_{a,b|a,a} + M_{a+1,b|a+1,a} + M_{a+1,b|a,a+1} = M_{a,b|a,a} + \sum M_{a+1,b|a+1,a}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(8) $b > c > a$

(8.1) $c > a + 1$ Let $\mu := \lambda + (1, 0, 0, 0)$.

$$P_\mu = M_{a+1,b|c,a} + M_{a+1,b|a,c} + M_{b,a+1|c,a} + M_{b,a+1|a,c}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_2) &= M_{a,b|c,a} + M_{a+1,b|c,a+1} + M_{a,b|a,c} + M_{a+1,b|a+1,c} + M_{b,a|c,a} + M_{b,a+1|c,a+1} \\ &\quad + M_{b,a|a,c} + M_{b,a+1|a+1,c} = \sum M_{a,b|c,a} + \sum M_{a+1,b|c,a+1} \end{aligned}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(8.2) $c = a + 1$ Let $\mu := \lambda + (0, 0, 0, 1)$.

$$P_\mu = M_{a,b|a+1,a+1} + M_{b,a|a+1,a+1}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_1) &= M_{a+1,b|a+1,a+1} + M_{a,b|a,a+1} + M_{a,b|a+1,a} + M_{b,a+1|a+1,a+1} + M_{b,a|a,a+1} + M_{b,a|a+1,a} \\ &= \sum M_{a,b|a+1,a} + \sum M_{a+1,b|a+1,a+1} \end{aligned}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(9) $b > c = a$

(9.1) $b > a + 1$ Let $\mu := \lambda + (1, 0, 0, 0)$.

$$P_\mu = M_{a+1,b|a,a} + M_{b,a+1|a,a}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_2) &= M_{a,b|a,a} + M_{a+1,b|a+1,a} + M_{a+1,b|a,a+1} + M_{b,a|a,a} + M_{b,a+1|a+1,a} + M_{b,a+1|a,a+1} \\ &= + \sum M_{a,b|a,a} + \sum M_{a+1,b|a+1,a} \end{aligned}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(9.2) $b = a + 1$ Let $\mu := \lambda + (1, 0, 0, 0)$.

$$P_\mu = M_{a+1,a+1|a,a}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_2) &= M_{a,a+1|a,a} + M_{a+1,a|a,a} + M_{a+1,a+1|a+1,a} + M_{a+1,a+1|a,a+1} \\ &= \sum M_{a,a+1|a,a} + \sum M_{a+1,a+1|a+1,a} \end{aligned}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(10) $c > b > a$

(10.1) $b > a + 1$ Let $\mu := \lambda + (1, 0, 0, 0)$.

$$P_\mu = M_{a+1,b|c,a} + M_{a+1,b|a,c} + M_{b,a+1|c,a} + M_{b,a+1|a,c}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_2) &= M_{a,b|c,a} + M_{a+1,b|c,a+1} + M_{a,b|a,c} + M_{a+1,b|a+1,c} \\ &\quad + M_{b,a|c,a} + M_{b,a+1|c,a+1} + M_{b,a|a,c} + M_{b,a+1|a+1,c} \\ &= \sum M_{a,b|c,a} + \sum M_{a+1,b|c,a+1} \end{aligned}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(10.2) $b = a + 1$ Let $\mu := \lambda + (0, 0, 0, 1)$.

$$P_\mu = M_{a+1,a+1|c,a} + M_{a+1,a+1|a,c}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_1) &= M_{a,a+1|c,a} + M_{a+1,a|c,a} + M_{a+1,a+1|c,a+1} + M_{a,a+1|a,c} + M_{a+1,a|a,c} + M_{a+1,a+1|a+1,c} \\ &= \sum M_{a,a+1|c,a} + \sum M_{a+1,a+1|c,a+1} \end{aligned}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(11) $c > b = a$

(11.1) $c > a + 1$ Let $\mu := \lambda + (0, 0, 0, 1)$.

$$P_\mu = M_{a,a|c,a+1} + M_{a,a|a+1,c}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_1) &= M_{a+1,a|c,a+1} + M_{a,a+1|c,a+1} + M_{a,a|c,a} + M_{a+1,a|a+1,c} + M_{a,a+1|a+1,c} + M_{a,a|a,c} \\ &= \sum M_{a,a|c,a} + \sum M_{a,a+1|c,a+1} \end{aligned}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(11.2) $c = a + 1$ Let $\mu := \lambda + (0, 0, 0, 1)$.

$$P_\mu = M_{a,a|a+1,a+1}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_1) &= M_{a+1,a|a+1,a+1} + M_{a,a+1|a+1,a+1} + M_{a,a|a,a+1} + M_{a,a|a+1,a} \\ &= \sum M_{a,a|a+1,a} + \sum M_{a,a+1|a+1,a+1} \end{aligned}$$

All items belong to P_λ by [Proposition 0.0.1](#).

□

Theorem 1.4. Let $a, b, c \in \mathbb{Z}$ and $\lambda = (a, b|a, c)$ with $a \neq b$ and $a \neq c$. The projective objects P_λ in $\mathfrak{B}_{b|c}$ have the following Verma flag formulae.

Proof. The Cases are as follows:

(1) $a > b > c$ Let $\mu := \lambda + (1, 0, 0, 0)$.

$$P_\mu = M_{a+1,b|a,c} + M_{a+1,b|c,a}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_2) &= M_{a,b|a,c} + M_{a+1,b|a+1,c} + M_{a,b|c,a} + M_{a+1,b|c,a+1} \\ &= \sum M_{a,b|a,c} + \sum M_{a+1,b|a+1,c} \end{aligned}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(2) $b > a > c$

(2.1) $b > a + 1$ Let $\mu := \lambda + (1, 0, 0, 0)$.

$$P_\mu = M_{a+1,b|a,c} + M_{a+1,b|c,a} + M_{b,a+1|a,c} + M_{b,a+1|c,a}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_2) &= M_{a,b|a,c} + M_{a+1,b|a+1,c} + M_{a,b|c,a} + M_{a+1,b|c,a+1} + M_{b,a|a,c} + M_{b,a+1|a+1,c} \\ &\quad + M_{b,a|c,a} + M_{b,a+1|c,a+1} = \sum M_{a,b|a,c} + \sum M_{a+1,b|a+1,c} \end{aligned}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(2.2) $b = a + 1$ Let $\mu := \lambda + (1, 0, 0, 0)$.

$$P_\mu = M_{a+1,a+1|a,c} + M_{a+1,a+1|c,a}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_2) &= M_{a,a+1|a,c} + M_{a+1,a|a,c} + M_{a+1,a+1|a+1,c} + M_{a,a+1|c,a} + M_{a+1,a|c,a} + M_{a+1,a+1|c,a+1} \\ &= \sum M_{a,a+1|a,c} + \sum M_{a+1,a+1|a+1,c} \end{aligned}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(3) $a > c > b$ Let $\mu := \lambda + (1, 0, 0, 0)$.

$$P_\mu = M_{a+1,b|a,c} + M_{a+1,b|c,a}$$

$$Pr_\lambda(P_\mu \otimes V_2) = M_{a,b|a,c} + M_{a+1,b|a+1,c} + M_{a,b|c,a} + M_{a+1,b|c,a+1} = \sum M_{a,b|a,c} + \sum M_{a+1,b|a+1,c}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(4) $c > a > b$

(4.1) $c > a + 1$ Let $\mu := \lambda + (0, 0, 1, 0)$.

$$P_\mu = M_{a,b|a+1,c}$$

$$Pr_\lambda(P_\mu \otimes V_1) = M_{a+1,b|a+1,c} + M_{a,b|a,c}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(4.2) $c = a + 1$ Let $\mu := \lambda + (0, 0, 1, 1)$.

$$P_\mu = M_{a,b|a+1,a+2}$$

$$Pr_\lambda(P_\mu \otimes V_3) = M_{a,b|a,a+1} + M_{a+1,b|a+1,a+1} + M_{a+2,b|a+1,a+2}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(5) $b > c > a$

(5.1) $c > a + 1$ Let $\mu := \lambda + (0, 0, 1, 0)$.

$$P_\mu = M_{a,b|a+1,c} + M_{b,a|a+1,c}$$

$$Pr_\lambda(P_\mu \otimes V_1) = M_{a+1,b|a+1,c} + M_{a,b|a,c} + M_{b,a+1|a+1,c} + M_{b,a|a,c} = \sum M_{a,b|a,c} + \sum M_{a+1,b|a+1,c}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(5.2) $c = a + 1, b > c + 1$ Let $\mu := \lambda + (0, 0, 1, 1)$.

$$P_\mu = M_{a,b|a+1,a+2} + M_{b,a|a+1,a+2}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_1) &= M_{a,b|a,a+1} + M_{a+1,b|a+1,a+1} + M_{a+2,b|a+1,a+2} \\ &\quad + M_{b,a|a,a+1} + M_{b,a+1|a+1,a+1} + M_{b,a+2|a+1,a+2} \\ &= \sum M_{a,b|a,a+1} + \sum M_{a+1,b|a+1,a+1} + \sum M_{a+2,b|a+1,a+2} \end{aligned}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(5.3) $c = a + 1, b = c + 1$ Let $\mu := \lambda + (0, 0, 1, 1)$.

$$P_\mu = M_{a+2,a+2|a,a+1}$$

$$\begin{aligned} Pr_\lambda(P_\mu \otimes V_1) &= M_{a+2,a+2|a+1,a+2} + M_{a+1,a+2|a+1,a+1} \\ &\quad + M_{a+2,a+1|a+1,a+1} + M_{a,a+2|a,a+1} + M_{a+2,a|a,a+1} \\ &= \sum M_{a,a+2|a,a+1} + \sum M_{a+1,a+2|a+1,a+1} + M_{a+2,a+2|a+1,a+2} \end{aligned}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(6) $c > b > a$

(6.1) $b > a + 1$ Let $\mu := \lambda + (1, 0, 0, 0)$.

$$P_\mu = M_{a+1,b|a,c} + M_{b,a+1|a,c}$$

$$Pr_\lambda(P_\mu \otimes V_2) = M_{a,b|a,c} + M_{a+1,b|a+1,c} + M_{b,a|a,c} + M_{b,a+1|a+1,c} = \sum M_{a,b|a,c} + \sum M_{a+1,b|a+1,c}$$

All items belong to P_λ by [Proposition 0.0.1](#).

(6.2) $b = a + 1$ Let $\mu := \lambda + (1, 0, 0, 0)$.

$$P_\mu = M_{a+1,a+1|a,c}$$

$$Pr_\lambda(P_\mu \otimes V_2) = M_{a,a+1|a,c} + M_{a+1,a|a,c} + M_{a+1,a+1|a+1,c} = \sum M_{a,a+1|a,c} + M_{a+1,a+1|a+1,c}$$

All items belong to P_λ by [Proposition 0.0.1](#).

□

2 $\mathfrak{gl}(3|2)$

2.1 degree of atypicality = 2

Theorem 2.1. *Let $a, b, c, d \in \mathbb{Z}$ and $\lambda = (a, b, c|b, c)$. The projective objects P_λ in $\mathfrak{B}_a$ have the following Verma flag formulae.*

Proof. The Cases are as follows:

(1) $a > b > c$

(1.1) $b > c + 1, a > b + 1$

(1.2) $b > c + 1, a = b + 1$

(1.3) $b = c + 1, a > b + 1$

(1.4) $b = c + 1, a = b + 1$

(2) $a > b = c$

(2.1) $a > b + 1$

(2.2) $a = b + 1$

(3) $a = b > c$

(3.1) $b > c + 1$

(3.2) $b = c + 1$

(4) $b > a > c$

(4.1) $a > c + 1$

(4.2) $a = c + 1, b > c + 1$

(4.3) $a = c + 1, b = c + 1$

(5) $b > a = c$

(5.1) $b > a + 1$

(5.2) $b = a + 1$

(6) $b = a > c$

(7) $a > c > b$

(7.1) $a > c + 1, c > b + 1$

(7.2) $a > c + 1, c = b + 1$

(7.3) $a = c + 1, c > b + 1$

(7.4) $a = c + 1, c = b + 1$

(8) $a > c = b$

(9) $a = c > b$

(9.1) $a > b + 1$

(9.2) $a = b + 1$

(10) $c > a > b$

(10.1) $a > b + 1$

$$(10.2) \quad c > a + 1, a = b + 1$$

$$(10.3) \quad c = a + 1, a = b + 1$$

$$(11) \quad c > a = b$$

$$(11.1) \quad c > b + 1$$

$$(11.2) \quad c = b + 1$$

$$(12) \quad c = a > b$$

$$(13) \quad b > c > a$$

$$(13.1) \quad b > c + 1$$

$$(13.2) \quad b = c + 1$$

$$(14) \quad b > c = a$$

$$(15) \quad b = c > a$$

$$(16) \quad c > b > a$$

$$(16.1) \quad c > b + 1$$

$$(16.2) \quad c = b + 1$$

$$(17) \quad c > b = a$$

$$(18) \quad c = b > a$$

$$(19) \quad c = b = a$$

□